

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION-BASED QUESTIONS

Code of Conduct:

I G. Dhanusha 3100 (192325012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Dhanusha

APPLICATION-BASED QUESTIONS

1. Scenario: Event Communication Security

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- Deduce the most likely original message.
- Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- Demonstrate how the encryption process transforms the message step-by-step.
- Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- Analyze how investigators could detect the presence of hidden data within the image.
- Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- Determine the most probable original message by reversing the encryption process.
- Critically evaluate why such a method is vulnerable in modern communication systems.
- Propose one practical enhancement to improve the confidentiality of messages in this scenario.

(2) Scenario: Intercepted Suspicious Message

Recover the most likely original message.

The given ciphertext is:

GHSRH RH Z HVXIVG

The message is encrypted using the Atbash cipher, a monoalphabetic substitution cipher in each letter is replaced by its reverse letter in the alphabet.

Atbash

Substitution

plain text

ciphertext

A
B
C
D
E
F
G
H
I
J
K
L
M

X
W
V
U
T
S
R
Q
P
O
N

Each letter is replaced with its corresponding reverse alphabet letter

Decryption :-

ciphertext:-

G R S R H R H Z H V X I V G

Plain text:-

T H I S I S A S ~~E~~ ~~E~~ R E T

Therefore, the original message is:-

THIS IS A SECRET

b)

Simple substitution:-

The Atbash cipher replaces each letter with its opposite letter in the alphabet without changing the length of the message.

2) Fixed mapping:-

Every letter has a fixed replacement

For ex:- G is the replacement by T, S by H, and R by I.

3> Word patterns:

The ciphertext contains four groups of characters : GSRH, R+I, Z and HVXI VGT.

4> NO secret key:

Atbash uses a fixed alphabet mapping and does not require a separate encryption key.

Q57 Scenario: Legacy Encryption in corporate communication:-

a) The given ciphertext is:

WECRLTEERDSD EEEFAOCAIYDEN

The message is encrypted using the Rail Fence cipher with 3 rails, a transposition cipher text that rearranges the positions of characters without changing the characters themselves.

Step 1:

Understand the encryption method

Step: 2

Arrange the plaintext in a

Zigzag pattern:-

WE ARE DISCOVERED FLEET AT ONCE

Step: 3

Read the characters row by row

Rail 1: WE CRRLTE (TOP row characters)

Rail 2: Middle row characters

Rail 3: Remaining bottom row characters.

Step: 4

Obtain the original message

WE ARE DISCOVERED FLEET AT ONCE

b)

* Small key space:-

The no. of possible rail arrangements is limited.

* Frequency analysis:-

Since the cipher only rearrange characters.

* No character substitutions:

The original letters are preserved.

* Brute force attack:-

An attacker can test multiple mail.

* Known - plaintext attack:-

If an attacker knows part of the original message.

* No Secret key

The method does not provide strong key based protection.

Therefore,

The rail fence cipher is not secure enough for protecting confidential business information in modern communication systems.

c)

A practical enhancement is to replace the rail fence cipher with AES (Advanced Encryption Standard) using GCM mode (AES-GCM)

Implementation :-

convert the original plaintext into bytes.

Generate a secure, random encryption key.

Encrypt the plaintext using AES-GCM.

Advantages of AES-GCM:-

Strong encryption

Authentication

Key based security

Modern security

Data integrity

The 3-rail fence cipher is vulnerable because it only rearranges characters and provides no strong protection against attacks.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation